

Symplectic Is Not Enough: Time-Reversal Symmetry Makes Learned Flow Maps Usable as Hamiltonian Monte Carlo Proposals

Abstract

Surrogate Hamiltonian Monte Carlo replaces an expensive log-density gradient with a learned approximation. Every such method we are aware of learns a *vector field* — a gradient, or an energy that is differentiated — and places it inside a symplectic integrator. This inherits the integrator’s correctness for free, but it also inherits its cost: advancing L leapfrog steps requires L sequential network evaluations. A learned *flow map* needs one, independently of L . We show why this substitution is not free, and what is required to make it sound. A Hamiltonian flow carries two structures: it preserves the symplectic form, and it is time-reversal symmetric. Symplectic networks encode the first by architecture and the second not at all — and it is the second that a Metropolis correction consumes, in the form of an involution. An unsymmetrized symplectic network therefore violates detailed balance by $\mathcal{O}(10^{-1})$, and no amount of training data repairs it. Composing the network with its momentum-conjugate inverse restores the missing symmetry exactly, as an algebraic identity in the weights and at no cost in parameters; the construction is that of Valperga et al. (2022), which we transfer from dynamics prediction to the sampling setting and instantiate with closed-form inverses for the LA- and G-SympNet layers. We further show that the usual all-pairs training set is not merely unnecessary but harmful: supervising only the single time offset the sampler queries is $1275\times$ cheaper at $L = 50$ and fits better. On Gaussian-process hyperparameter inference at the benchmark configuration of Li et al. (2019), the resulting sampler attains $3.7\times$ the effective samples per second of exact HMC, while every integrator-based surrogate we test is *slower* than exact HMC. We also report a cautionary result: an unsymmetrized network posts a competitive 142 ESS/s at a reversibility error of 5×10^{-4} , illustrating that effective sample size alone cannot certify a surrogate sampler.

Keywords: symplectic neural networks, Hamiltonian Monte Carlo, time-reversal symmetry, structure-preserving architectures, surrogate models

1. Introduction

Structure-preserving architectures encode a known property of a physical system into a hypothesis class, so that it holds for every choice of weights rather than being approached through training. Symplectic neural networks (SympNets; Jin et al., 2020) are a clean instance: each layer is a shear on phase space, so the network is symplectic by construction and its Jacobian has unit determinant identically. They sit in a broader effort to learn Hamiltonian dynamics with architectural constraints (Greydanus et al., 2019; Chen et al., 2019).

Hamiltonian dynamics, however, carry more than one structure. Writing $z = (q, p)$ and $R(q, p) = (q, -p)$ for the momentum flip, the exact flow Φ_t of a Hamiltonian even in the momentum satisfies both

$$\Phi_t^*(dq \wedge dp) = dq \wedge dp \quad \text{and} \quad R \circ \Phi_t \circ R = \Phi_t^{-1}. \quad (1)$$

The first is symplecticity; the second is time-reversal symmetry. A SympNet inherits the first from its architecture. It does not inherit the second, and the usual training objective — a regression of predicted onto observed states — does not reward it.

We study the setting where this omission is not a matter of accuracy but of correctness, and where it also has a concrete payoff: surrogate Hamiltonian Monte Carlo.

Why one would want a learned flow map. Hamiltonian Monte Carlo (Brooks et al., 2011) proposes states by integrating Hamiltonian dynamics and corrects the discretization error with a Metropolis step. When the target’s gradient is expensive, surrogate methods learn a cheap stand-in during warm-up: a gradient (Li et al., 2019), or a Hamiltonian that is differentiated (Dhulipala et al., 2023). In every such method the learned object is a *vector field* placed inside a symplectic integrator. That inheritance is why they remain asymptotically exact — an inaccurate network lowers the acceptance rate but cannot bias the chain — but it also fixes their cost: advancing L leapfrog steps costs L sequential network evaluations.

A learned flow map is different in exactly this respect. Because $\hat{\phi}(z, \tau)$ takes the elapsed time as an input, a proposal costs one evaluation regardless of L . The saving grows with trajectory length, which is precisely the regime adaptive schemes such as NUTS (Hoffman and Gelman, 2011) and ChEES (Hoffman et al., 2021) push towards. The obstacle is that replacing the integrator forfeits the correctness it was supplying, and the network must now provide that structure itself.

Contributions.

- We show that a SympNet used as an MCMC proposal satisfies volume preservation but violates involutivity, by $\mathcal{O}(10^{-1})$ at initialization and up to $\mathcal{O}(10^0)$ after training (Table 2). The resulting sampler is not asymptotically exact, and additional data does not fix it.
- We transfer the reversible-symplectic construction of Valperga et al. (2022) from dynamics prediction to sampling, and instantiate it with closed-form inverses for the LA- and G-SympNet layers, making it exact rather than approximate. We restate why it satisfies both Metropolis conditions (Proposition 1) and is consistent (Lemma 2).
- We show that the standard all-pairs training set for flow maps is actively harmful: supervising only the offset the sampler queries is $1275\times$ cheaper at $L = 50$ and yields *higher* effective sample size (Table 1).
- Empirically, the symmetrized flow map attains $3.7\times$ the ESS/s of exact HMC on GP hyperparameter inference, where every integrator-based surrogate is slower than exact HMC (Figure 1a), with the advantage growing in L (Figure 1b).

2. Background

2.1. Validity of a deterministic Metropolis proposal

Let $\pi(z) \propto e^{-H(z)}$ on $\mathbb{R}^{2D}$ with $H \circ R = H$. A Metropolis–Hastings scheme proposing $z' = T(z)$ deterministically and accepting with probability $\min\{1, e^{H(z)-H(Tz)}\}$ leaves π

invariant provided

$$(V) \quad |\det \nabla T(z)| = 1, \quad (2)$$

$$(I) \quad (R \circ T) \circ (R \circ T) = \text{Id}, \quad (3)$$

volume preservation and involutivity of $R \circ T$ (Green, 1995; Tierney, 1998). Leapfrog integration of any position-dependent force satisfies both identically. That neural MCMC proposals must be engineered to satisfy them is established (Song et al., 2017; Levy et al., 2018), and Spanbauer et al. (2020) give a general construction from volume-preserving involutions. Our observation is that the SympNet family, adopted precisely for its structural guarantees, supplies (2) and not (3).

2.2. Symplectic networks

A SympNet composes layers that update either q or p using a function of the other, unmodified half of the state; for the gradient module in “up” mode, $(q, p) \mapsto (q, p + \tau K^\top a \sigma(Kq + b))$. Each layer has unit-determinant Jacobian, so (2) holds for any weights. Condition (3) concerns how the map interacts with R , and no layer constrains it.

3. Symmetrizing the flow map

3.1. Closed-form inverses

The shear structure that makes SympNets symplectic also makes them exactly invertible. In an “up” layer the position passes through untouched, so the increment added to p can be recomputed from the output and subtracted; the inverse is the same expression with the sign reversed, at identical cost and with no iteration. The LA module applies a shear followed by a translation, so its inverse subtracts the translation first and then undoes the shears in reverse order. Composing layer inverses in reverse order gives Φ_t^{-1} exactly; we measure $\|\Phi_t^{-1}\Phi_t z - z\|_\infty \approx 10^{-15}$ in double precision. Valperga et al. (2022) build their reversible architecture from Hénon maps and coupling layers; the SympNet modules admit the same treatment, which lets the construction be applied to networks already in use for Hamiltonian learning.

3.2. The symmetrized map

Following Valperga et al. (2022), define

$$\Psi_t := R \circ \Phi_{t/2}^{-1} \circ R \circ \Phi_{t/2}. \quad (4)$$

Proposition 1 *If $\Phi_{t/2}$ is invertible then Ψ_t satisfies (3); if it is additionally symplectic then Ψ_t satisfies (2). The Metropolis–Hastings chain with proposal Ψ_t therefore leaves π invariant for arbitrary network weights.*

Proof Write $\Phi := \Phi_{t/2}$. Since $R^2 = \text{Id}$, $R \circ \Psi_t = \Phi^{-1} \circ R \circ \Phi$, a conjugation of the involution R , hence an involution: $(R\Psi_t)^2 = \Phi^{-1}R\Phi\Phi^{-1}R\Phi = \Phi^{-1}R^2\Phi = \text{Id}$, which is (3). For (2), R is anti-symplectic, so $R\Phi^{-1}R$ is symplectic and Ψ_t , a composition of two

Table 1: Pair construction for the symmetrized flow map (Gaussian target). Supervising only the queried offset is dramatically cheaper *and* fits better as L grows.

L	pairs	per traj.	total	train (s)	ESS
5	all	15	9 000	10.0	762
5	from x_0	5	3 000	7.8	763
5	endpoint	1	600	7.7	709
25	all	325	195 000	72.0	352
25	from x_0	25	15 000	23.3	348
25	endpoint	1	600	20.1	1175
50	all	1275	765 000	72.4	429
50	from x_0	50	30 000	37.3	267
50	endpoint	1	600	36.3	711

symplectic maps, is symplectic; hence $|\det \nabla \Psi_t| = 1$. Neither argument uses any property of Φ beyond invertibility and symplecticity. ■

Lemma 2 *If Φ_s is the exact time- s flow of a Hamiltonian with $H \circ R = H$, then $\Psi_t = \Phi_t$.*

Proof By (1), $\Phi_{t/2}^{-1} = R\Phi_{t/2}R$; substituting into (4) and using $R^2 = \text{Id}$ gives $\Psi_t = \Phi_{t/2} \circ \Phi_{t/2} = \Phi_t$. ■

Equation (4) is the adjoint (symmetrization) composition of geometric integration (Hairer et al., 2006), with the adjoint realized as $R \circ \Phi^{-1} \circ R$. The property survives composition: if $R\Psi R = \Psi^{-1}$ then $R\Psi^k R = \Psi^{-k}$, so $(R\Psi^k)^2 = \text{Id}$ for every k , and multi-step trajectories remain valid proposals. The map costs two network evaluations rather than one and adds no parameters; note that a Ψ built from an n -block network has the same effective depth as a plain $2n$ -block network, so comparisons must control for effective depth rather than block count.

3.3. What the sampler actually queries

Flow maps are conventionally trained on all pairs (x_i, x_j) , $i < j$, drawn from each warm-up trajectory, giving $\mathcal{O}(L^2)$ examples per trajectory. But the sampler evaluates $\hat{\phi}$ at exactly one offset, $\tau = L\epsilon$. The extra offsets are not free: with finite capacity, fitting times the sampler never queries takes capacity from the one it does. Table 1 compares supervising all pairs, all offsets from x_0 , and only (x_0, x_L) .

4. Experiments

Models are trained on warm-up trajectories from exact HMC and used as proposals, corrected with the exact Hamiltonian. All models share an optimization protocol (Adam at 10^{-2} , learning rate halved on plateau, early stopping, best-iterate restoration), so comparisons reflect architecture rather than tuning. Timings are measured on an isolated GPU.

Table 2: Validity conditions at random initialization ($D = 2$, $L = 5$, $\epsilon = 0.1$, double precision). Volume preservation is $\max |\det \nabla T| - 1$; involutivity is $\max \|(RT)^2 z - z\|_\infty$.

proposal map	$ \det \nabla T - 1$	involution error	valid
Exact leapfrog (HMC)	$2 \cdot 10^{-16}$	$3 \cdot 10^{-16}$	yes
Learned force + leapfrog	$3 \cdot 10^{-16}$	$7 \cdot 10^{-16}$	yes
Learned separable H + leapfrog	$7 \cdot 10^{-16}$	$1 \cdot 10^{-15}$	yes
Learned field + leapfrog	$8 \cdot 10^{-16}$	$9 \cdot 10^{-16}$	yes
LA-SympNet (plain)	$1 \cdot 10^{-15}$	$2.4 \cdot 10^{-1}$	no
G-SympNet (plain)	$7 \cdot 10^{-16}$	$1.2 \cdot 10^{-1}$	no
LA-SympNet (symmetrized)	$2 \cdot 10^{-15}$	$4 \cdot 10^{-16}$	yes
G-SympNet (symmetrized)	$7 \cdot 10^{-16}$	$1 \cdot 10^{-16}$	yes

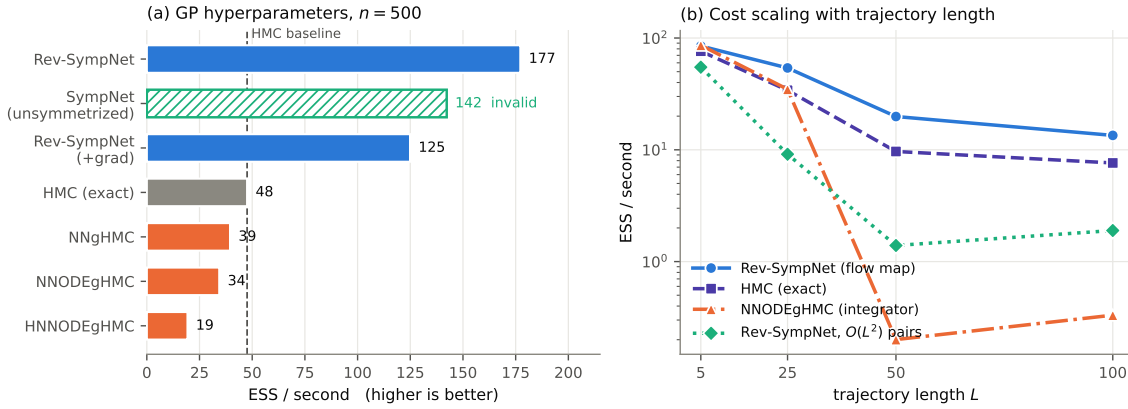


Figure 1: (a) Gaussian-process hyperparameter inference at the configuration of [Li et al. \(2019\)](#) ($n = 500$, Matérn $\nu = 3/2$, $\epsilon = 0.05$, $L = 20$, 10^4 samples); best value over training budgets. The symmetrized flow map attains $3.7\times$ the ESS/s of exact HMC, while every integrator-based surrogate is slower than exact HMC. The unsymmetrized network (hatched) posts a competitive number at a reversibility error of 5×10^{-4} and is not a valid sampler. (b) ESS/s against trajectory length. The flow map’s advantage grows with L ; the integrator-based surrogate collapses because it compounds error over L steps; and training on $\mathcal{O}(L^2)$ pairs erases the gain.

Structural validity. Table 2 evaluates (2) and (3) directly at random initialization, emphasizing that these are properties of the construction and not of training. Integrator-based surrogates satisfy both to machine precision. Plain SympNets satisfy volume preservation and violate involutivity by $\mathcal{O}(10^{-1})$. The symmetrized versions satisfy both.

Expensive-likelihood target. We use GP hyperparameter inference at the configuration of Li et al. (2019): $n = 500$ observations with four features, a Matérn kernel with $\nu = 3/2$, two sampled hyperparameters, $\epsilon = 0.05$, $L = 20$, 10^4 samples. Each gradient requires an $\mathcal{O}(n^3)$ Cholesky factorization. Figure 1a reports the best ESS/s over training budgets. The symmetrized flow map reaches 177 ESS/s against exact HMC’s 48, a factor of 3.7; the gain is wall-clock, not sample quality, since HMC attains higher raw ESS (18 634 versus 11 609) in $6\times$ the time. Every integrator-based surrogate is slower than exact HMC ($0.41\text{--}0.83\times$).

Scaling with trajectory length. Figure 1b varies L on a Gaussian target. The symmetrized flow map exceeds exact HMC at every L , by $2.05\times$ at $L = 50$. The integrator-based surrogate degrades sharply — its ESS falls to 20.5 at $L = 50$ — because it applies its learned field L times and compounds error at each step, whereas the flow map applies once. Training on all pairs removes the advantage entirely, consistent with Table 1.

Effective sample size does not certify a surrogate. In Figure 1a the unsymmetrized network attains 142 ESS/s, which would read as a $3\times$ improvement over exact HMC, at a reversibility error of 5×10^{-4} . A map with a large involution defect can produce weakly autocorrelated paths while targeting the wrong distribution; ESS measures autocorrelation, not correctness. Structural diagnostics of the kind in Table 2 should accompany any reported ESS for a learned proposal. We note this because the artifact is easy to mistake for a result.

5. Discussion

The observation we wish to emphasize is structural. When an architecture is called structure-preserving it is worth asking which structures, and whether the intended use requires others. SympNets preserve the symplectic form exactly and are adopted for that reason, but the flows they model are also time-reversal symmetric, and it is that second symmetry an accept–reject correction consumes. Because it can be imposed exactly by composition, at no cost in parameters and with no loss of consistency, there is little reason to leave it to training.

Two limitations bound our claims. First, our targets are low-dimensional (two to thirty parameters); the flow map must represent a map on the full $2D$ -dimensional phase space, whereas gradient-learning surrogates fit a field on the D -dimensional position space alone, and we would expect that gap to widen with dimension. Second, our speedups are measured on GPU, where a 500×500 Cholesky costs about 4 ms and the gradient is not the dominant per-step cost; Li et al. (2019) report a $39.3\times$ speedup for a gradient-learning surrogate on their hardware, which we do not reproduce ($0.83\times$) at matched n and L . Our advantage comes from a different mechanism — collapsing L sequential operations into one — and the two numbers are not comparable.

Finally, supervising only the queried offset makes the map fast and accurate but fixes the trajectory length at training time. Adaptive-length schemes (Hoffman and Gelman, 2011; Hoffman et al., 2021) would require the $\mathcal{O}(L)$ construction, at a measurable cost in fit; making trajectory length adaptive for a flow map — where, unlike for an integrator, length carries no per-proposal cost — is the natural next question.

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